

Ryan Strawbridge

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EDUCATION

Northeastern University – Boston, MA

May 2027

Bachelor of Science in Mechanical Engineering (BSME)

GPA: 3.58

Coursework: Heat Transfer | Fluid Mechanics | Solid Mechanics | Thermodynamics | Material Science | Dynamics

Activities: AeroNU Propulsion Member | Running Club President | Rhythm Guitarist in Alt. Rock Band

Achievements: DoD Security Clearance | CSWA Certified | Lean Six Sigma Certification | 800th in the Boston Marathon

PROFESSIONAL EXPERIENCE

Commonwealth Fusion Systems

Devens, MA

Manufacturing / Equipment Engineering Co-op (Term 2)

Jan 2026 – Jun 2026

- Designed and validated a 2,100 lb hand-portable lifting frame to **BTH-1** and **ASME B30.20**, earning sign-off from structural engineering and safety; increased table space for fixture tooling during magnet stacking and **saved 30–45 minutes per lift**
- Developed a modular CNC seam-cutting fixture accommodating 11 cable configurations, enabling qualification testing ahead of a new production line; **applied GD&T** with position and symmetry controls on dowel-pin datums for repeatable alignment, **reducing future operation time by over 80%**
- Deployed a low-cost adjustable fixture using welded tabs and off-the-shelf linear rails and carriages to maintain cable alignment during insulation wrapping, improving process robustness and **reducing labor by 0.5–3 hours per magnet layer**

Manufacturing Engineering Co-op (Term 1)

Jan 2025 – Jun 2025

- Re-designed a hydraulic piston-driven bending machine in Siemens NX using bolt safety analysis and first-principles calculations, cutting operation time by 75% and reducing injury risk; **still in use over a year later, saving hundreds of labor hours**
- Developed a predictive equation relating bend offset, bend radius, and arc length through iterative spring-back experiments and data interpolation, **eliminating trial-and-error die design** and enabling future cable-bending dies to be sized analytically on the first iteration
- Owned a work cell** and wrote detailed work instructions with technicians covering cable assembly, joint placement, orbital cutting, and welding, replacing tribal knowledge with a repeatable reference used on the production floor

Raytheon Missiles & Defense

Andover, MA

Operations Engineering Co-op

Jan 2024 – Sep 2024

- Prototyped a composite fixture using SolidWorks and PCB Gerber files to shield heat-sensitive filters from 93 °C while enabling reflow of adjacent components, **increasing first-pass yield** and removing filter damage as a scrap driver on the line
- Ran a nine-configuration solder stencil study varying placement and volume to provide off-gas escape channels, reducing voiding and **removing a rework step** from the process
- Co-owned a work cell** and developed thermal reflow profiles by refining oven parameters through thermocouple data, operator input, and X-ray inspection to IPC-J-STD-001 Class 3, **reducing rework time across 15 programs**

PROJECTS

Underwater Acoustic Release Mechanism – Capstone Project

Boston, MA

Mechanical Design Lead – Release System

Jun 2026 – Present

- Own the mechanical design** of a low-cost underwater acoustic release mechanism for underwater robot recovery; designing a rotary lever arm mechanism and using first-principles force analysis to evaluate and rule out initial design concepts
- Performed **structural analysis** of the release shaft under combined axial compression and torsion, and selected O-ring seal geometry using Parker's O-Ring Handbook to keep the rotating, pressure-sealed shaft interface leak-free at 300 m depth
- Validating hand-calculated stress predictions on housing and latch components in **Ansys FEA** under combined pressure and mechanical loading

AerospaceNU – Liquid Rocketry Club

Boston, MA

Liquid Rocket Engine Member

Sep 2023 – Present

- Designing and validating an actively controlled nitrogen pressurant subsystem, integrating a servo-actuated ball valve and torque coupling with closed-loop pressure regulation based on downstream pressure feedback
- Supported setup and execution of **three successful static-fire tests**, integrating propulsion, electrical, and instrumentation systems while configuring test hardware and data acquisition equipment

TECHNICAL SKILLS

Software: Siemens NX | SolidWorks | Ansys Mechanical | Arduino | Teamcenter | ManufacturO | Odoo | Confluence | Jira

Design: Structural Analysis, DFM, GD&T, FEA (Ansys – developing)

Fabrication: Manual G-Code | CNC Milling | Power Tools | TIG Welding | 3D Printing | Soldering | Water Jetting | Laser Cutting